



Software Process Improvement –SEPI-471

Lean, SE Wastes, Value Add Vs Non Value Add

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Agenda

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Agenda

- Lean
- Wastes
- VA Vs NVA

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Lean Software development

- is a software development methodology that
 - focuses on delivering value to the customer by eliminating waste, improving efficiency, and continuously improving the development process.
- It is inspired by the principles of Lean Manufacturing developed by the Toyota production system.

Lean Software development

- Lean principles originally came from manufacturing, especially the Toyota Production System.
- Later these ideas were applied to software development and formalized in the book:
- Lean Software Development: An Agile Toolkit by Mary Poppendieck and Tom Poppendieck.

7 Lean Software Development Principles

- Eliminate Waste**

- Remove anything that doesn't add value to the customer,
 - such as unnecessary features, rework, delays, and inefficient communication.

- Build Quality In**

- Focus on preventing defects rather than fixing them later.
- Use automated testing, continuous integration, and peer reviews to maintain high-quality code.

- Create Knowledge**

- Encourage continuous learning and experimentation. Use feedback loops,
 - documentation, and knowledge-sharing to improve processes.

- Defer Commitment –**

- Avoid making irreversible decisions too early.
- Keep options open and make informed choices based on the latest available data.

7 Lean Software Development Principles

- Deliver Fast**

- Reduce cycle time by focusing on small, incremental releases. Frequent deliveries allow
- faster feedback and adaptation to changing requirements.

- Empower team**

- Foster a culture of collaboration and trust.
- encourage open communication, and recognize individual contributions.

- Optimize the complete Picture**

- Improve the entire software development process rather than just individual parts.
- Consider end-to-end efficiency, from requirements gathering to deployment.

Waste

non-value-adding
activities

activities that
absorb resources
and increase cost
without adding
value

activities and work
products that do not
contribute to
customer value

Impediment to
Flow

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Software Engineering Waste

Waiting

Defect

Rework

Mental Distress

Needless Composite Solutions

Motion

Knowledge Loss

Ineffective Communication

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Software Engineering Wastes

Waiting



- Delay in a process
- **Causes**
 - lack of knowledge sharing
 - lax or untrustworthy testing or code reviews
 - omitted required information
 - unavailability of human and technical resources

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Software Engineering Wastes

Defect



- an instance in which a requirement is not satisfied
- **Causes**
 - inexperienced developers
 - poor coding strategies
 - inappropriate code review strategies

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Software Engineering Wastes

Rework



- Defect correction process
- **Causes**
 - inadequate quality assurance
 - violation of standards
 - poor review strategies

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Software Engineering Wastes

Mental Distress



- **feeling of being unable to cope with mental or emotional pressure**
- **Causes**
 - Unnecessary psychological stress
 - Cognitive load (e.g., due to poor source code)

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Software Engineering Wastes

Needless Composite Solution



- Performing task in unnecessary complex manner
 - Complex user interface, complex code solution
- **Causes**
 - Lack of reuse strategy

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Software Engineering Wastes

Motion



- Amount of effort of the developer to find the answer to the problem
- **Causes**
 - Unavailability of needed information
 - Lack of communication

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Software Engineering Wastes

Knowledge Loss



- Cost of re-acquiring knowledge once team knew
- **Causes**
 - Team Churn
 - Knowledge Silos (Lack of KS)

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Software Engineering Wastes

In-Effective Communication



- Incomplete, incorrect, misleading, or absence of communication
- **Causes**
 - Large Team Size
 - Imbalance Communication (Dominating the conversation, not listening)
 - Inefficient Meetings

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Software Engineering Waste-generalized

Waiting

Defect

Rework

Mental Distress

Needless
Composite
Solutions

Motion

Knowledge Loss

Ineffective
Communication



Waste identification and elimination

Software Engineering Perspectives

- Software Requirement Engineering
- Software Modelling
- Software Development
- Software Testing
- Software Deployment

Educational system

Any business process

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Value Adding Steps/Activities

Value-add steps must meet three criteria

1. The step must change the form or function of the product or service.
2. The customer must be willing to pay for the change.
3. The step must be performed correctly first time.

Examples:

Development, shipping a product, packaging the product, receiving the order, invoice creation.

Non-Value Adding Steps/Activities

- 1) A step that does not affect the form or function of the product or service.
- 2) The customer is not willing to pay for the change.
- 3) The step includes rework.

Examples: Peer reviewing, checking, moving product, quality assessments.

Non-Value Adding Steps/Activities Which are Necessary

Non value-add steps which are necessary refers to those steps which:

- 1) Are legal requirements.
- 2) Are governance requirements
- 3) Are company requirements.

Examples: Send information within 24 hours due to California State law, sign off by governance committee required.



Thank you

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